



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO MARTHAR AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

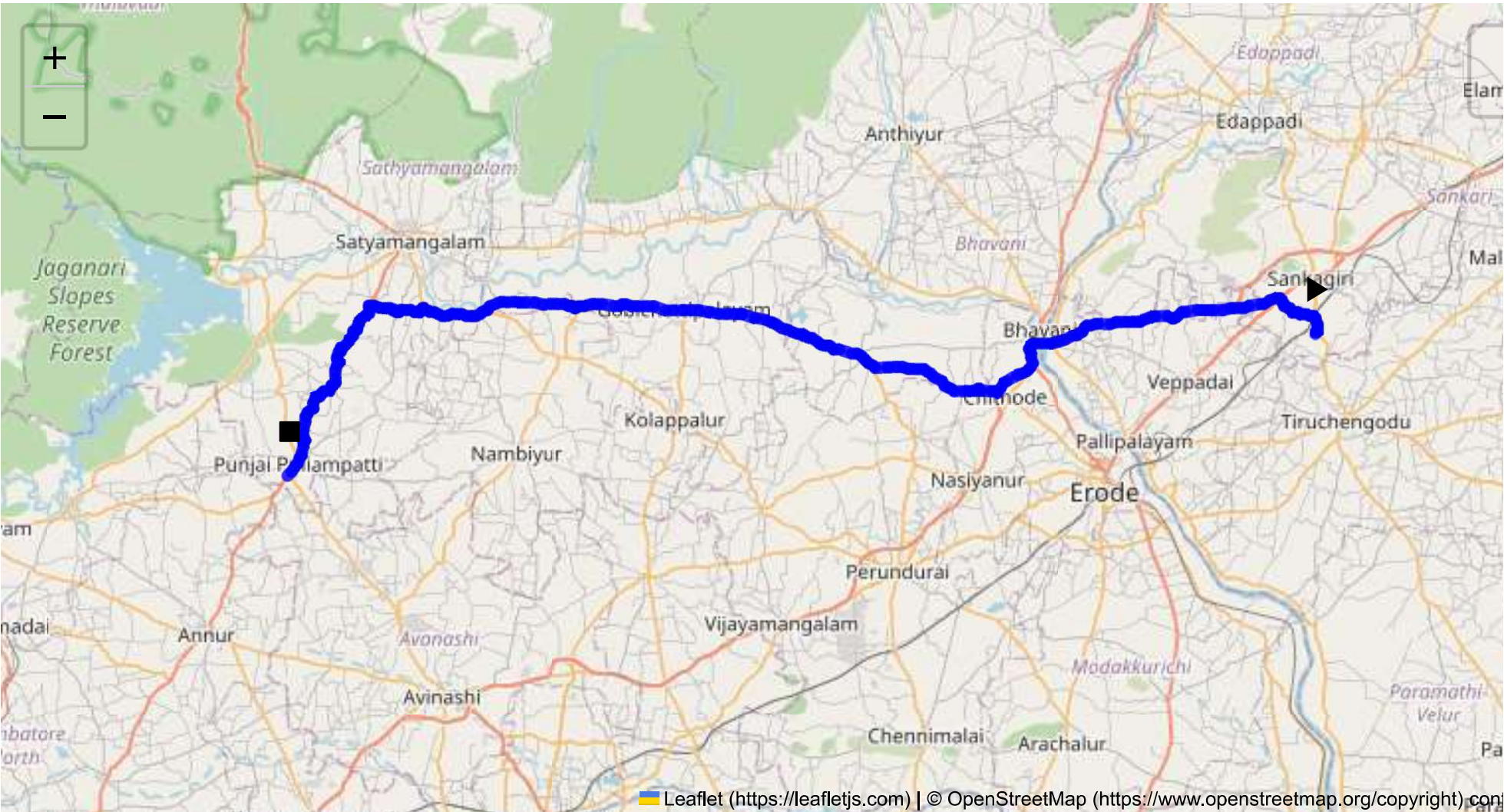
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 95.48 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.4 hours
Start: (11.4381, 77.8734)
End: (11.341931, 77.160114)



Welcome to the Journey Risk Management Study

To perform a comprehensive analysis of the route from CVQF+23W, Sangagiri to 85R6+P2G, Pulampatti, Tamil Nadu, I have broken down the route analysis into the specified points.

1. Overview of the Route Map

- The route predominantly follows the state highways, namely SH 86 and SH 188, with a minor detour on rural roads near Pulampatti. It traverses through towns like Salem, Bhavani, and areas

in proximity to the Bhavani River.

2. Typical Weather Conditions and Potential Weather-Related Hazards

- This region experiences a tropical savannah climate with hot summers [March-June], heavy monsoon rains [October-December], and mild winters. Monsoons pose a risk of flooding and waterlogging, affecting road conditions and visibility.

3. Analysis of Traffic Patterns

- High traffic density is typical during morning (8-10 AM) and evening (5-8 PM) due to local commuter traffic, particularly near Salem and Erode regions. Congestion is likely near industrial zones and market areas.

4. Assessment of Road Quality and Infrastructure

- While state highways are generally well-maintained, occasional patches of potholes and uneven road surfaces can be expected, especially post-monsoon. Rural roads may have narrower lanes and lack proper lighting.

5. Suggestions for Alternative Routes for Emergencies

- In emergencies, NH544 (part of the former NH47, a more extensive, four-lane national highway) may serve as an alternative, though detours and additional distance should be anticipated.

6. Summary of Local Regulations Affecting Hazardous Material Transport

- Permits required for transporting hazardous materials must be obtained for each district crossed. Restrictions on transport timings and prohibited zones near populated areas, schools, and places of worship should be adhered to.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- While specific historical data is limited, regional trends suggest occasional spillage incidents due to vehicle overturns or accidents in congested area. Precautionary measures must be emphasized to prevent such incidents.

8. Environmental Considerations and Sensitive Areas

- Proximity to Bhavani River and adjacent agriculture lands necessitates care to prevent contamination from spills. Areas with dense flora and fauna also ensure minimal environmental disruptions.

9. Analysis of Communication Coverage

- Some rural segments may experience patchy mobile network coverage, particularly in valleys or densely vegetated areas which could impact communication during emergencies.

10. Estimated Emergency Response Times

- Urban sections within Salem and Erode may see emergency response within 20-30 minutes. Remote rural areas might face delays up to 60 minutes due to accessibility issues.

11. Overall Summary of Risk Assessment

- The route poses moderate risk mainly due to weather, congestion, and variable road conditions. Proper planning is critical, emphasizing route familiarization, rigid adherence to safety protocols,

and having contingency plans for communication or route alterations. Regular updates on weather forecasts and traffic advisories can mitigate risk.

In conclusion, diligent adherence to local regulations and ongoing communication with logistical operations centers can highly minimize risks associated with transporting hazardous materials along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.44965, 77.77565	30 KM/Hr
8	Turn	Medium	11.44953, 77.77344	30 KM/Hr
9	Turn	Medium	11.43165, 77.67734	30 KM/Hr
10	Turn	High	11.39829, 77.65233	15 KM/Hr
11	Turn	Medium	11.45763, 77.36309	30 KM/Hr
12	Turn	Medium	11.45750, 77.35617	30 KM/Hr
13	Turn	Medium	11.45536, 77.25423	30 KM/Hr
14	Turn	High	11.45616, 77.25388	15 KM/Hr
15	Turn	High	11.45630, 77.25276	15 KM/Hr
16	Turn	High	11.45413, 77.25226	15 KM/Hr
17	Turn	High	11.45371, 77.25149	15 KM/Hr
18	Turn	High	11.45298, 77.25182	15 KM/Hr
19	Turn	High	11.45270, 77.25157	15 KM/Hr
20	Turn	High	11.45322, 77.24737	15 KM/Hr
21	Turn	Medium	11.45376, 77.24707	30 KM/Hr
22	Turn	High	11.45308, 77.23644	15 KM/Hr
23	Turn	Medium	11.45545, 77.23047	30 KM/Hr
24	Turn	High	11.45516, 77.22964	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Turn	High	11.45580, 77.22882	15 KM/Hr
26	Turn	Medium	11.45555, 77.22843	30 KM/Hr
27	Turn	High	11.45708, 77.21660	15 KM/Hr
28	Turn	Medium	11.45099, 77.21508	30 KM/Hr
29	Turn	Medium	11.42096, 77.19395	30 KM/Hr
30	Turn	Medium	11.41965, 77.19493	30 KM/Hr
31	Turn	Medium	11.40475, 77.19260	30 KM/Hr
32	Turn	Medium	11.40421, 77.19221	30 KM/Hr
33	Turn	High	11.40325, 77.19002	15 KM/Hr
34	Turn	Medium	11.40005, 77.18895	30 KM/Hr
35	Turn	Medium	11.39979, 77.18857	30 KM/Hr
36	Turn	Medium	11.39934, 77.18451	30 KM/Hr
37	Turn	Medium	11.39736, 77.18344	30 KM/Hr
38	Turn	Medium	11.39588, 77.18182	30 KM/Hr
39	Turn	Medium	11.39574, 77.18080	30 KM/Hr
40	Turn	Medium	11.39053, 77.17784	30 KM/Hr
41	Turn	Medium	11.38749, 77.17806	30 KM/Hr
42	Turn	Medium	11.38315, 77.17283	30 KM/Hr
43	Turn	Medium	11.35462, 77.16942	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Dhanvanthri Hospital	11.4496712, 77.7593772	30 km/h	Medium
4	hospital	J.K.K. Trust Hospital	11.4445841, 77.7307962	30 km/h	Medium
6	hospital	Shri Sathyanarayana Hospital	11.4291297, 77.6913408	30 km/h	Medium
7	hospital	Thanish Siddha Hospital	11.430003, 77.674964	30 km/h	Medium
8	clinic	Harshitha Clinic	11.4313207, 77.674718	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam	11.432369, 77.674894	30 km/h	Medium
10	clinic	G.K Clinic	11.4297244, 77.6749715	30 km/h	Medium
11	hospital	Naarayani Hospital	11.4157401, 77.5773087	30 km/h	Medium
12	hospital	P K Hospital	11.4175923, 77.5640962	30 km/h	Medium
13	hospital	Dharun Hospital	11.4223638, 77.5597735	30 km/h	Medium
14	hospital	Dr. Ramaswamy Hospital	11.4249065, 77.5535658	30 km/h	Medium
15	hospital	Dakshin Trauma Centre	11.454593, 77.4461819	30 km/h	Medium
16	hospital	Sri Varahi Herbal Research Centre and Hospital	11.4539333, 77.4427438	30 km/h	Medium
17	hospital	Dakshin Hospital	11.4546301, 77.445905	30 km/h	Medium
18	clinic	Germen Homeopathy Clinic	11.4536305, 77.4394692	30 km/h	Medium
19	hospital	JP Hospital	11.454765, 77.438257	30 km/h	Medium
20	hospital	KKS Hospital	11.4551792, 77.4392927	30 km/h	Medium
22	hospital	Siranjeeve Hospitals	11.453132, 77.4344382	30 km/h	Medium
23	clinic	Anandhan Clinic	11.4548037, 77.4349074	30 km/h	Medium
24	hospital	Ramachandran Hospital	11.452242, 77.434641	30 km/h	Medium
25	hospital	Kumaran Hospital	11.4533971, 77.4349621	30 km/h	Medium
26	hospital	Dr. Rathinasamy Hospital	11.457562, 77.4327752	30 km/h	Medium
27	hospital	Gobi Neuro Mind Care	11.4548344, 77.434649	30 km/h	Medium
28	hospital	Hariharan Hospital	11.4523578, 77.4299287	30 km/h	Medium
29	hospital	Vasanth Hospital	11.4526864, 77.4292173	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
30	clinic	Jeeva Nursing Home	11.4533031, 77.4294145	30 km/h	Medium
31	hospital	Sathya Hospital	11.4561856, 77.4298694	30 km/h	Medium
32	hospital	S K Hospital	11.4540106, 77.430108	30 km/h	Medium
33	hospital	Sivakami Hospital	11.4531846, 77.4299288	30 km/h	Medium
34	clinic	Sri Sudha Physiotherapy Clinic	11.4559915, 77.4300217	30 km/h	Medium
35	hospital	Best Hospital	11.4543558, 77.4263617	30 km/h	Medium
36	clinic	Anand Nursing Home	11.4553169, 77.4296248	30 km/h	Medium
37	clinic	Kanmani Child Care Clinic	11.4545273, 77.428668	30 km/h	Medium
38	hospital	Rajasekaran Hospital	11.454463, 77.4262	30 km/h	Medium
39	hospital	Samuel Hospital	11.4542555, 77.4271194	30 km/h	Medium
40	hospital	Sivasakthi Hospital	11.4544711, 77.4297229	30 km/h	Medium
41	hospital	SS Children's Hospital	11.4555386, 77.4296357	30 km/h	Medium
42	clinic	Sri Dhanvantari Siddha Clinic	11.4561216, 77.4295797	30 km/h	Medium
43	hospital	Neha Hospital	11.4560138, 77.4270219	30 km/h	Medium
44	hospital	M.S Children's Hospital	11.4556012, 77.4301006	30 km/h	Medium
45	hospital	Gobi Diabetic Hospital	11.455394, 77.4301181	30 km/h	Medium
46	clinic	Dr. Ramesh Babu Clinic	11.4560871, 77.4296353	30 km/h	Medium
47	hospital	Sankareswari Eye Hospital	11.4572997, 77.428765	30 km/h	Medium
48	hospital	Arun Hospital	11.4548852, 77.4250314	30 km/h	Medium
51	hospital	Government Hospital	11.4565126, 77.4288768	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
52	hospital	Sri Chakra Hospital	11.4544464, 77.4228363	30 km/h	Medium
53	hospital	U M Hospital	11.4544151, 77.4240386	30 km/h	Medium
54	hospital	Dr. Gopal Hospital	11.455258, 77.422901	30 km/h	Medium
55	clinic	VG Ortho Centre	11.4550504, 77.4225102	30 km/h	Medium
56	clinic	Palaniappa Clinic	11.4555925, 77.4202828	30 km/h	Medium
57	hospital	Abi SK Hospital	11.457163, 77.412487	30 km/h	Medium
58	clinic	Nalam Vazh S.K Siddha and Varma Clinic	11.457576, 77.412164	30 km/h	Medium
60	clinic	Singiripalayam Natarajan Siddha Ortho Care	11.4603252, 77.3183801	30 km/h	Medium
62	hospital	Government Hospital	11.3560512, 77.1689477	30 km/h	Medium
63	hospital	Arunachalam Hospital	11.3546609, 77.1698426	30 km/h	Medium
64	clinic	Mani Nursing Home	11.3525705, 77.1662763	30 km/h	Medium
65	clinic	Sri Saravana Clinic	11.351058, 77.1670823	30 km/h	Medium
66	hospital	Muthu Hospital	11.3512958, 77.1671183	30 km/h	Medium
67	hospital	Vishnu Nursing Home	11.351926, 77.165558	30 km/h	Medium
68	hospital	Thangam Health Care Hospital	11.3484244, 77.1652678	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
1	college	Vivekanandha Engineering College	11.4589312, 77.7899284	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
2	marketplace	Monday market	11.452863, 77.775989	30 km/h	Medium
5	school	SSM Matriculation Higher Secondary School	11.4321653, 77.6880046	30 km/h	Medium
21	school	Shree Vidyalaya Matric Higher Secondary School	11.4536114, 77.4383163	30 km/h	Medium
49	school	Diamond Jubilee Higher Secondary School	11.4529541, 77.4280306	30 km/h	Medium
50	school	CKK Palaniammal School	11.4569211, 77.4276828	30 km/h	Medium
59	school	Saratha Techno School	11.4544819, 77.379986	30 km/h	Medium
61	school	Saratha International School	11.4541471, 77.3032962	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44965, 77.77565



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44953, 77.77344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43165, 77.67734



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39829, 77.65233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45763, 77.36309



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45750, 77.35617



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45536, 77.25423



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45616, 77.25388



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45630, 77.25276



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45413, 77.25226



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45371, 77.25149



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45298, 77.25182



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45270, 77.25157



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45322, 77.24737



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45376, 77.24707



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45308, 77.23644



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.23047



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45516, 77.22964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45580, 77.22882



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45555, 77.22843



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45708, 77.21660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45099, 77.21508



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42096, 77.19395



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41965, 77.19493



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40475, 77.19260



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40421, 77.19221



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40325, 77.19002



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40005, 77.18895



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39979, 77.18857



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39934, 77.18451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39736, 77.18344



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39588, 77.18182



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39574, 77.18080



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39053, 77.17784



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38749, 77.17806



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38315, 77.17283

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.35462, 77.16942